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Studierichting: Informatica

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$f(x, y, z) = \sqrt{xyz}$ in $a(2, -1, -2)$ in de richting $h(1, 2, -2)$

$$f(2, -1, -2) = 2$$

$$\begin{aligned} Df(a, h) &= \lim_{\lambda \rightarrow 0} \frac{f((2, -1, -2) + \lambda(1, 2, -2)) - 2}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\sqrt{-4\lambda^3 - 10\lambda^2 - 2\lambda + 4} - 2}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{-4\lambda^3 - 10\lambda^2 - 2\lambda + 4 - 4}{\lambda \cdot (\sqrt{-4\lambda^3 - 10\lambda^2 - 2\lambda + 4} + 2)} \\ &= \lim_{\lambda \rightarrow 0} \frac{-4\lambda^2 - 10\lambda - 2}{\sqrt{-4\lambda^3 - 10\lambda^2 - 2\lambda + 4} + 2} \\ &= -\frac{1}{2} \end{aligned}$$

References